COMPUTER SIMULATION

Experiment #3: AC Measurements

Lab Section:

Section 13

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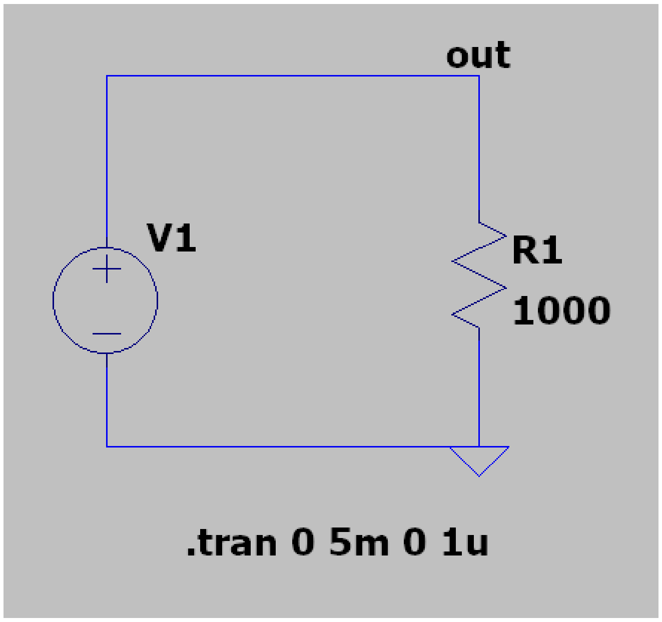
We were told to verify our results obtained in "PREPARATION" using a simulation of LTspice. In this simulation, Using a 1kHz, 10 V_{pp} sine wave as an example, perform the following steps in LTspice. We must enter the appropriate start time, stop time, and maximum timestep to obtain a waveform with 5 cycles. The three different types of waves we're obtaining are the sinusoidal wave, a square wave and a triangle wave.



To the left is the standard model for the AC circuit that we are basing our waveforms off of. Along with the necessary parameters.

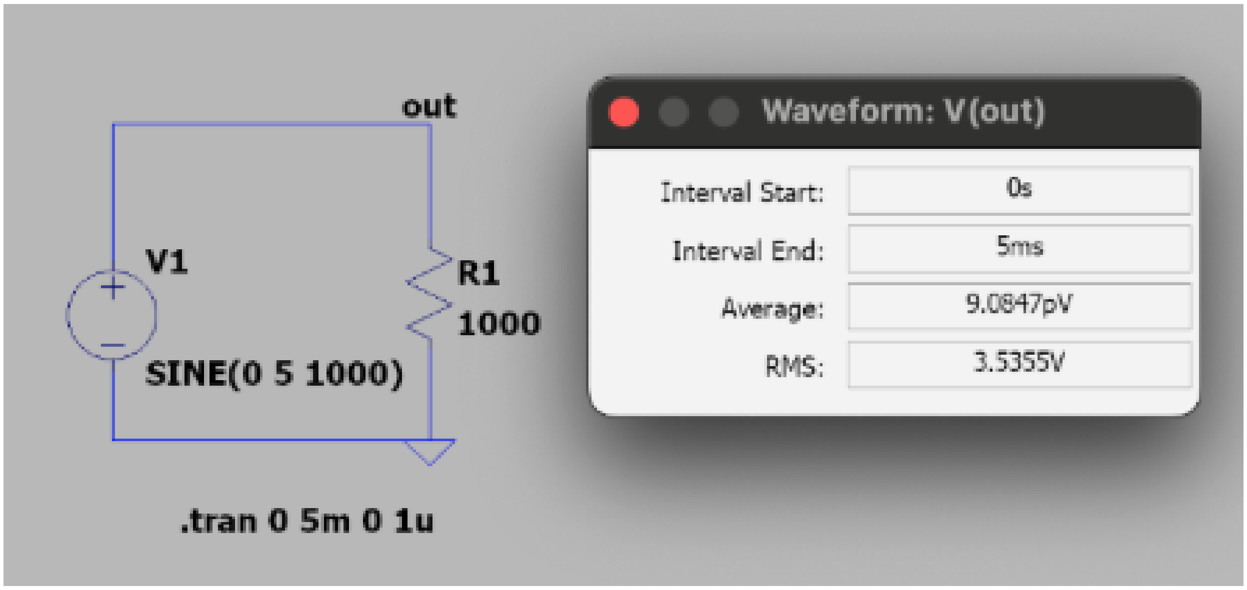
Under V1, is where the different voltage sources are inputted to receive the type of wave we desire. For the sine wave, the voltage source should be, sine(0 5 1000) where 0 = DC offset, 5 = amplitude, and 1000 = frequency.

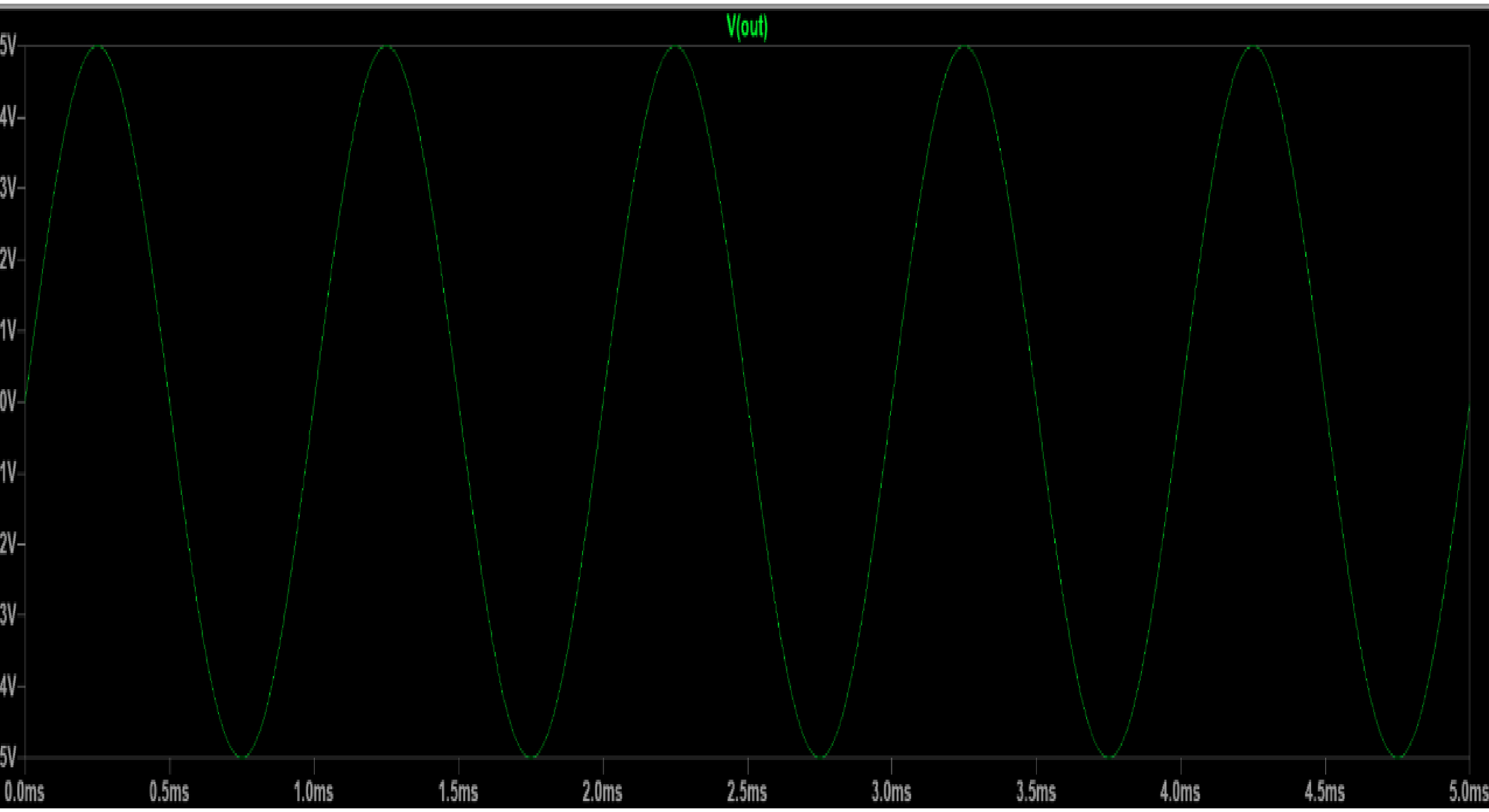
For the square wave, the voltage source should be pulse(-5 5 0 1n 1n 0.5m 1m). And for the triangle wave, pulse(-5 5 0 0.5m 0.5m 1n 1m). Using these, we can run our simulation with the above circuit, and test the voltage of the node labeled out, and see the waveform completed.

We use -5V and 5V for our range since our Vpp = 10V and we can also determine the period of our waveform given the frequency being 1kHz = (1kHz)^-1 = 1ms.

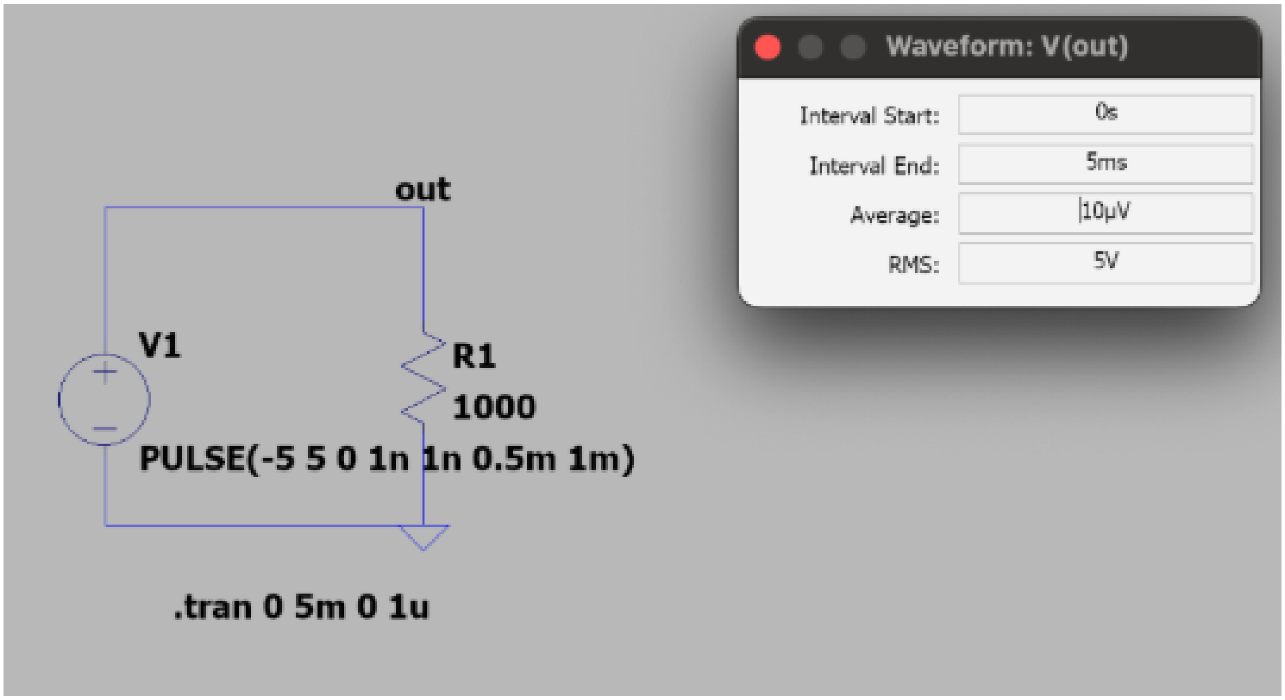
The simulation was completed and the results were recorded, screenshotted and shown below.

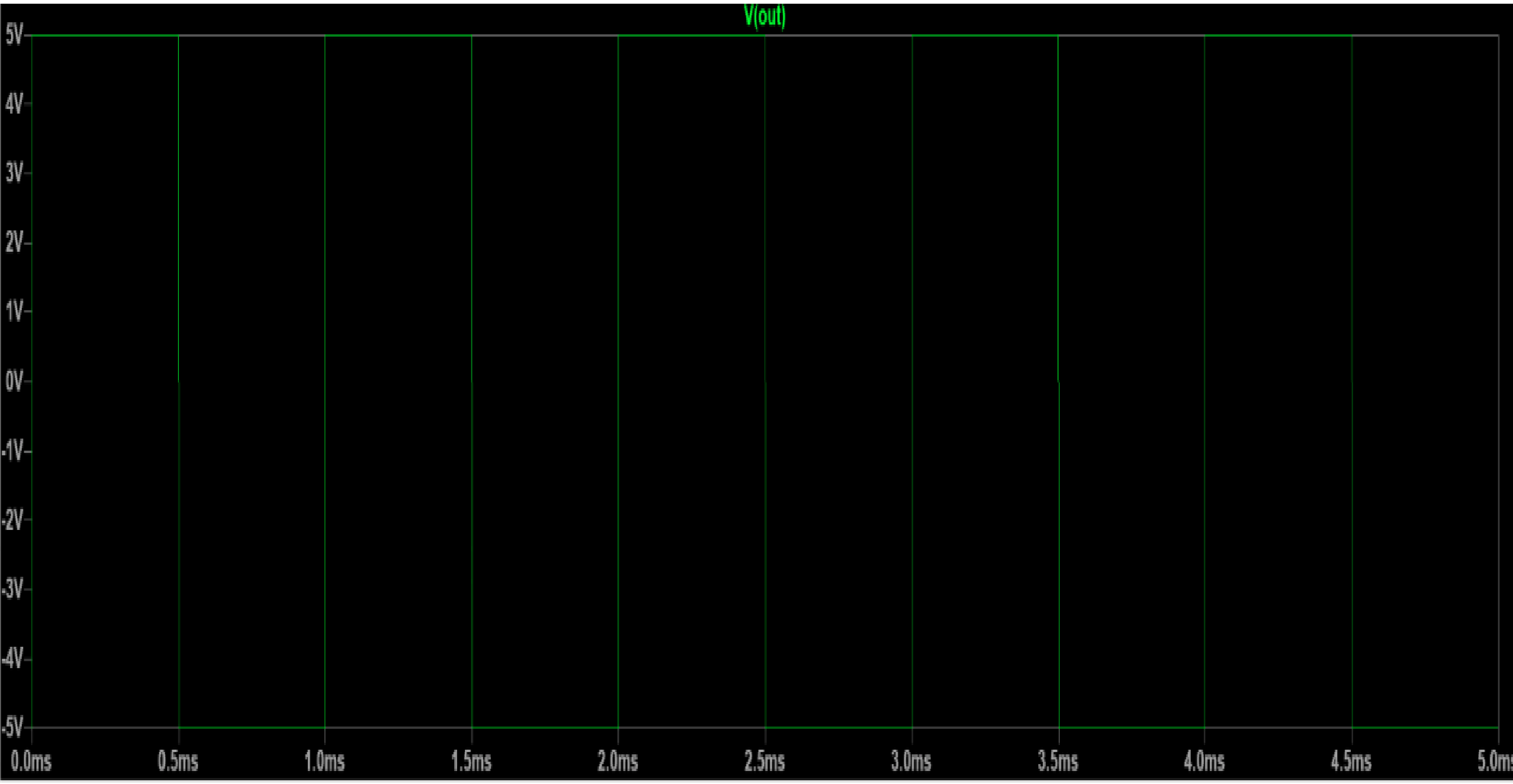
Sine Wave:


out
V1
R1
1000
SINE(0 5 1000)
.tran 0 5m 0 1u
Waveform: V(out)
Interval Start: 0s
Interval End: 5ms
Average: 9.0847pV
RMS: 3.5355V

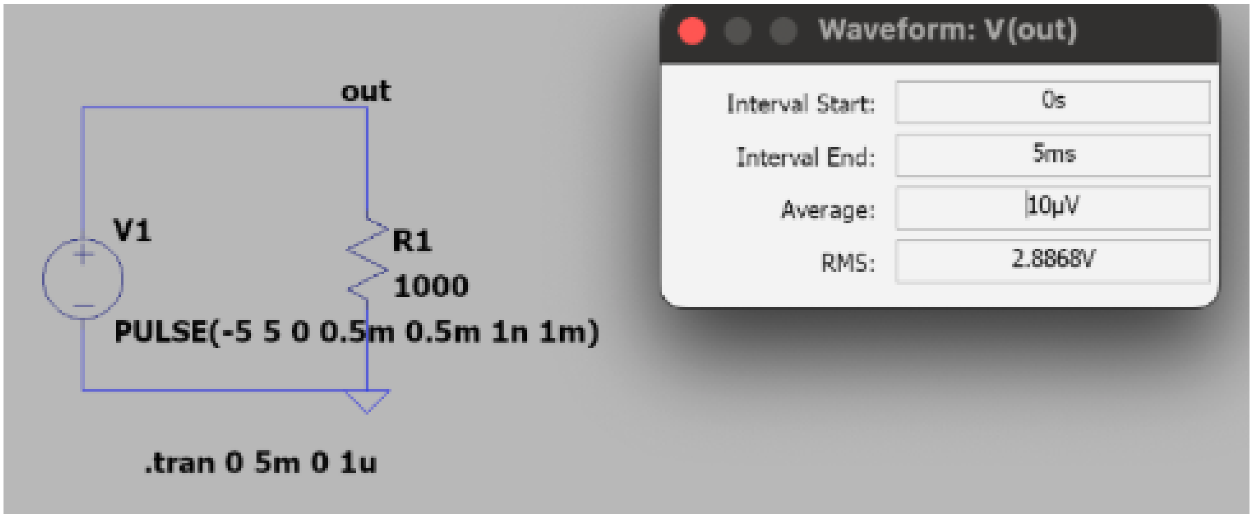

V(out)
5V
4V
3V
2V
1V
0V
1V
2V
3V
4V
5V
0.0ms
0.5ms
1.0ms
1.5ms
2.0ms
2.5ms
3.0ms
3.5ms
4.0ms
4.5ms
5.0ms

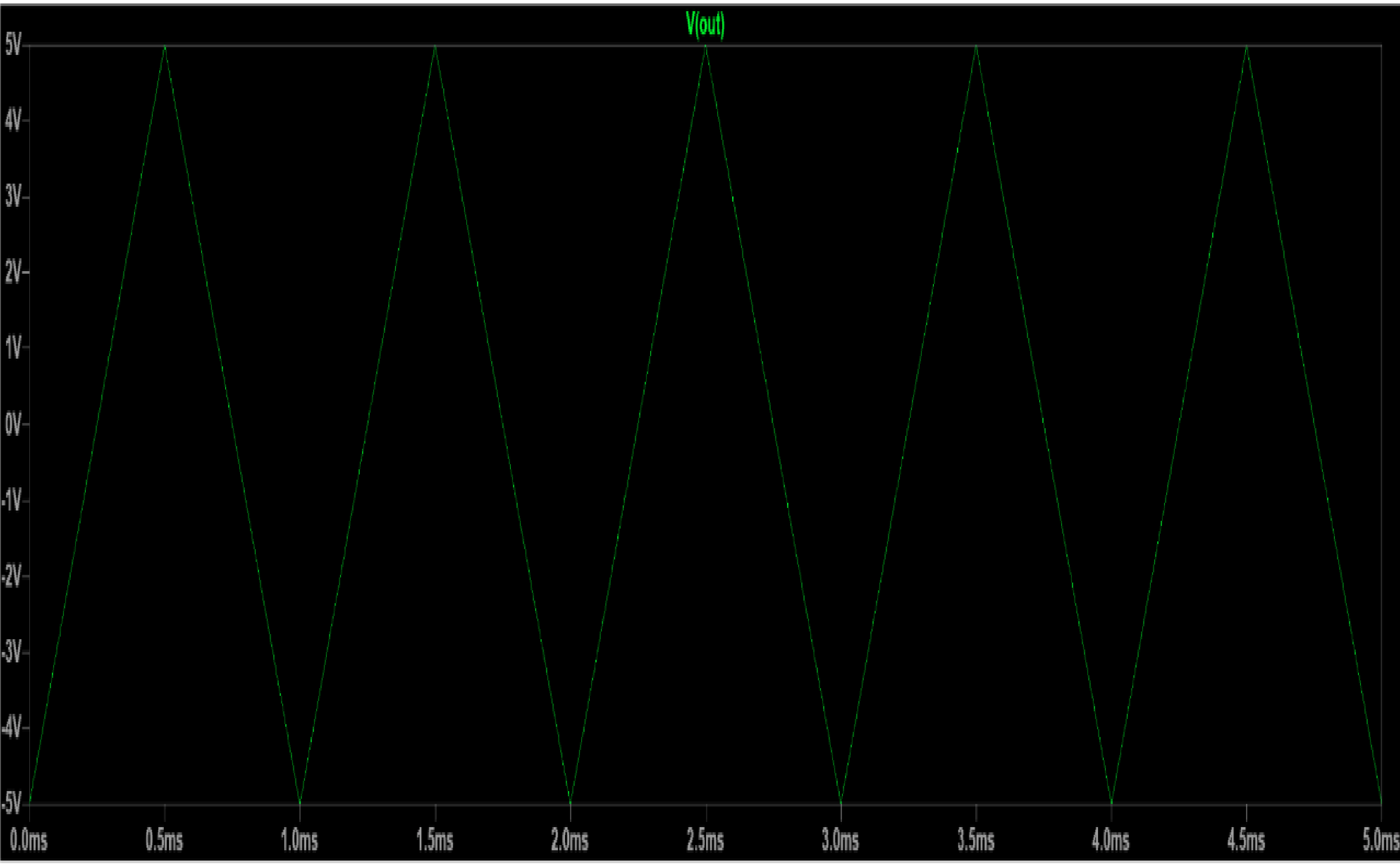
Square Wave:


Waveform: V(out)
Interval Start: 0s
Interval End: 5ms
Average: 10µV
RMS: 5V
out
V1
R1
1000
PULSE(-5 5 0 1n 1n 0.5m 1m)
.tran 0 5m 0 1u


V(out)
5V
4V
3V
2V
1V
0V
-1V
-2V
-3V
-4V
-5V
0.0ms
0.5ms
1.0ms
1.5ms
2.0ms
2.5ms
3.0ms
3.5ms
4.0ms
4.5ms
5.0ms

Triangle Wave:





Simulation Results

Waveform	**Sim. V. Source**	**Sim vs Hand V(RMS)**
Sine	sine(0 5 1000)	3.54 V = $5/\sqrt{2}$ V
Square	pulse(-5 5 0 1n 1n 0.5m 1m)	5 V = 5 V
Triangle	pulse(-5 5 0 0.5m 0.5m 1n 1m)	2.89 V = $5/\sqrt{3}$ V

These directly match up with our hand analysis.

Answers and results from simulation were cross checked and reviewed in comparison to hand analysis results and computation to learn, construct and finalize.